



Folium: Small-Scale Native App for Community Citizen Science and Botanical Cataloging

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INDUSTRIAL PROJECT

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1 Problem & Objectives



Context & Problem

- Bounded locations — parks, reserves, campuses, urban corridors — host **significant botanical biodiversity** with scarce structured digital documentation.
- iNaturalist** and **Flora Catalana** offer no offer no location-specific scoping, role hierarchies, or community-tailored gamification.
- No **mobile-first, community-driven, georeferenced** platform existed for local botanical monitoring.



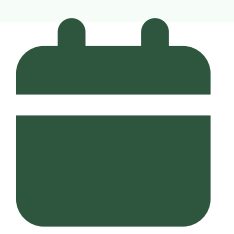
Objectives

- Native **mobile app** for submitting georeferenced citizen-science observations.
- Weighted-consensus **validation system** for botanical data quality assurance.
- GDPR** compliance and **production deployment** (Google Play Store).
- UAB Campus as a pilot "**Living Lab**" for hyper-local biodiversity monitoring.

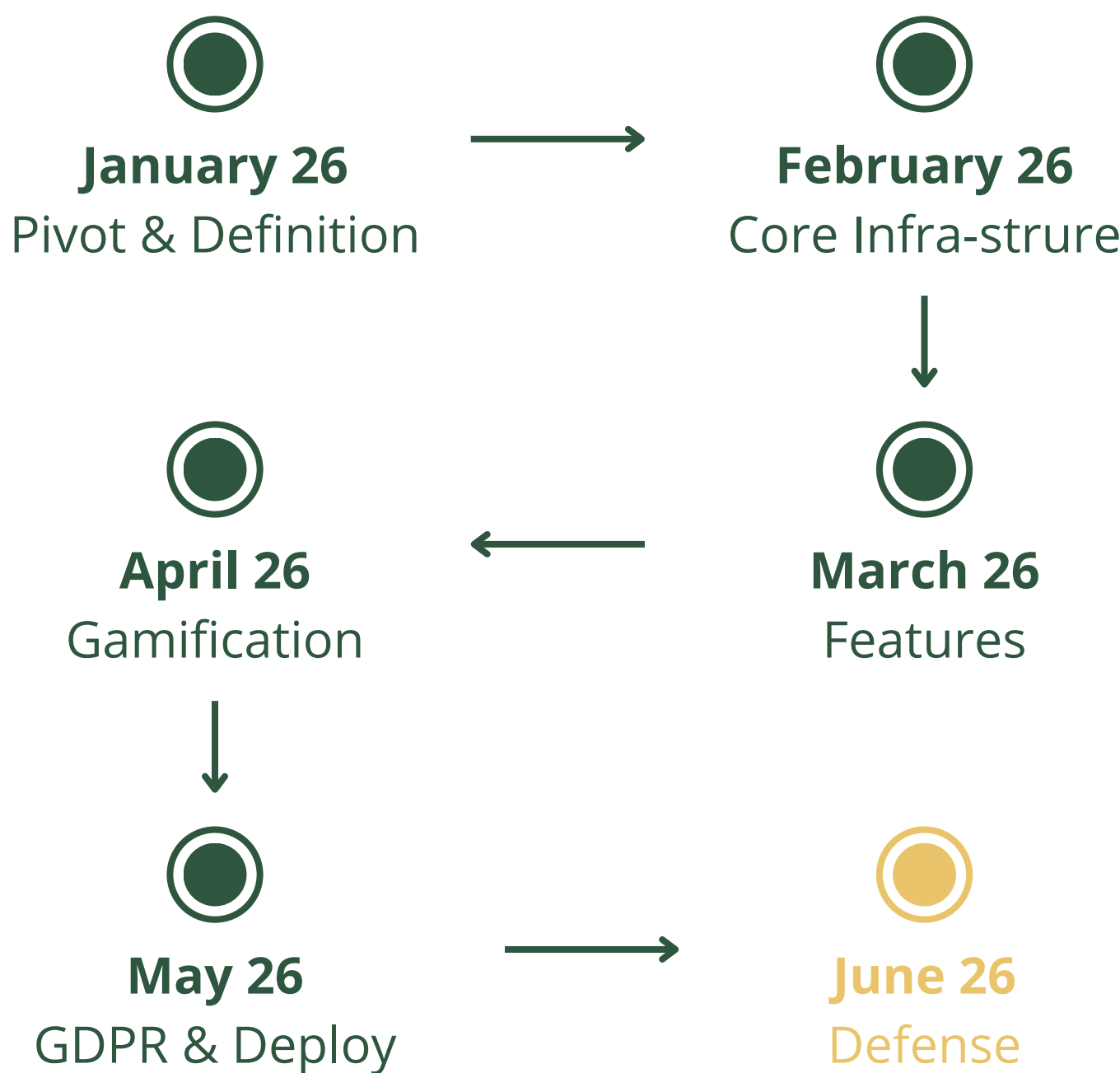


Benchmarks

- iNaturalist** — Consensus validation model (Research Grade $\geq 2/3$ agreement)
- ICO** (Catalan Ornithology Institute) — Expert alert system for anomalous observations
- Creative Commons BY-NC 4.0** — Licensing framework for all user-generated content



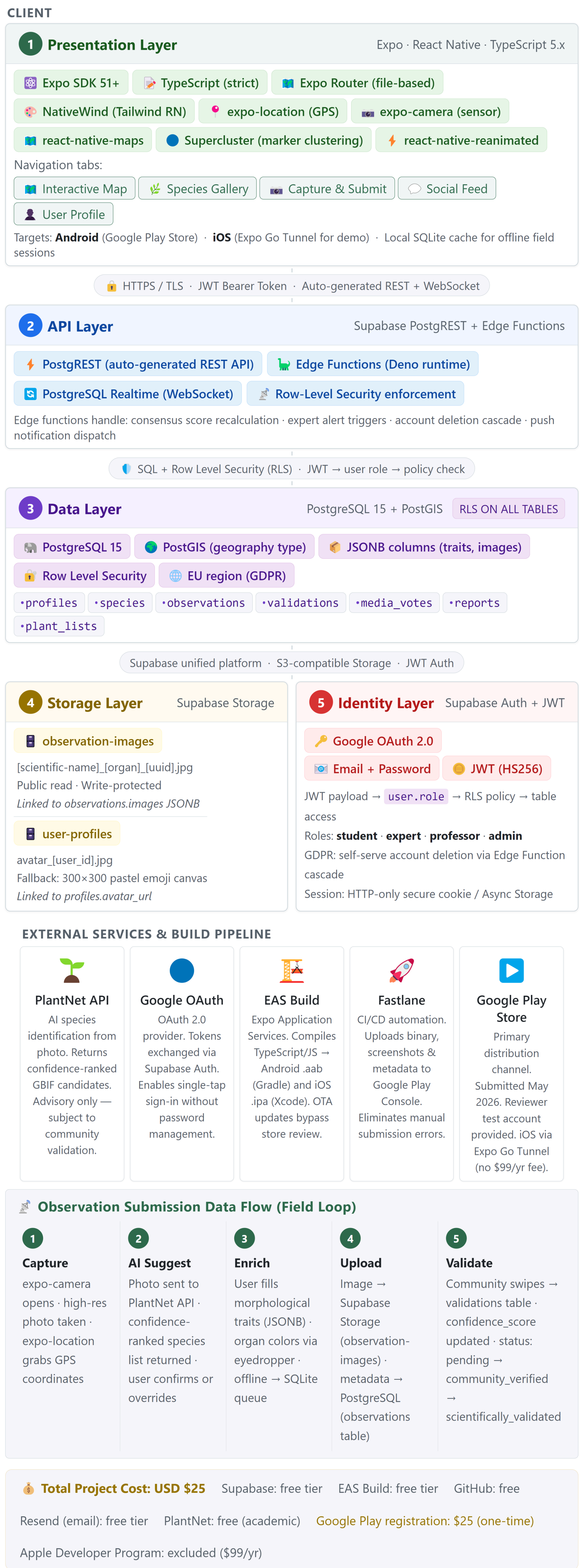
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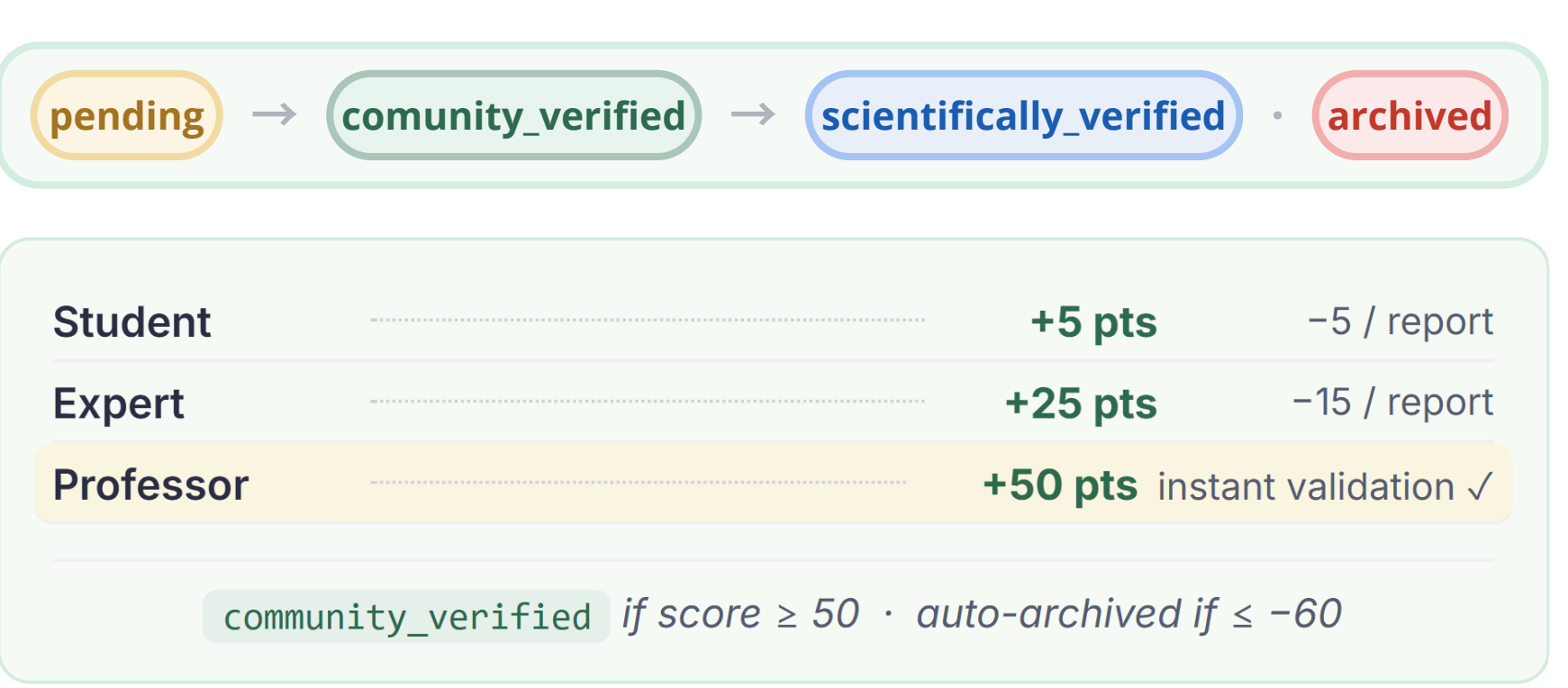
2 Architecture & Methodology



Architecture Diagram



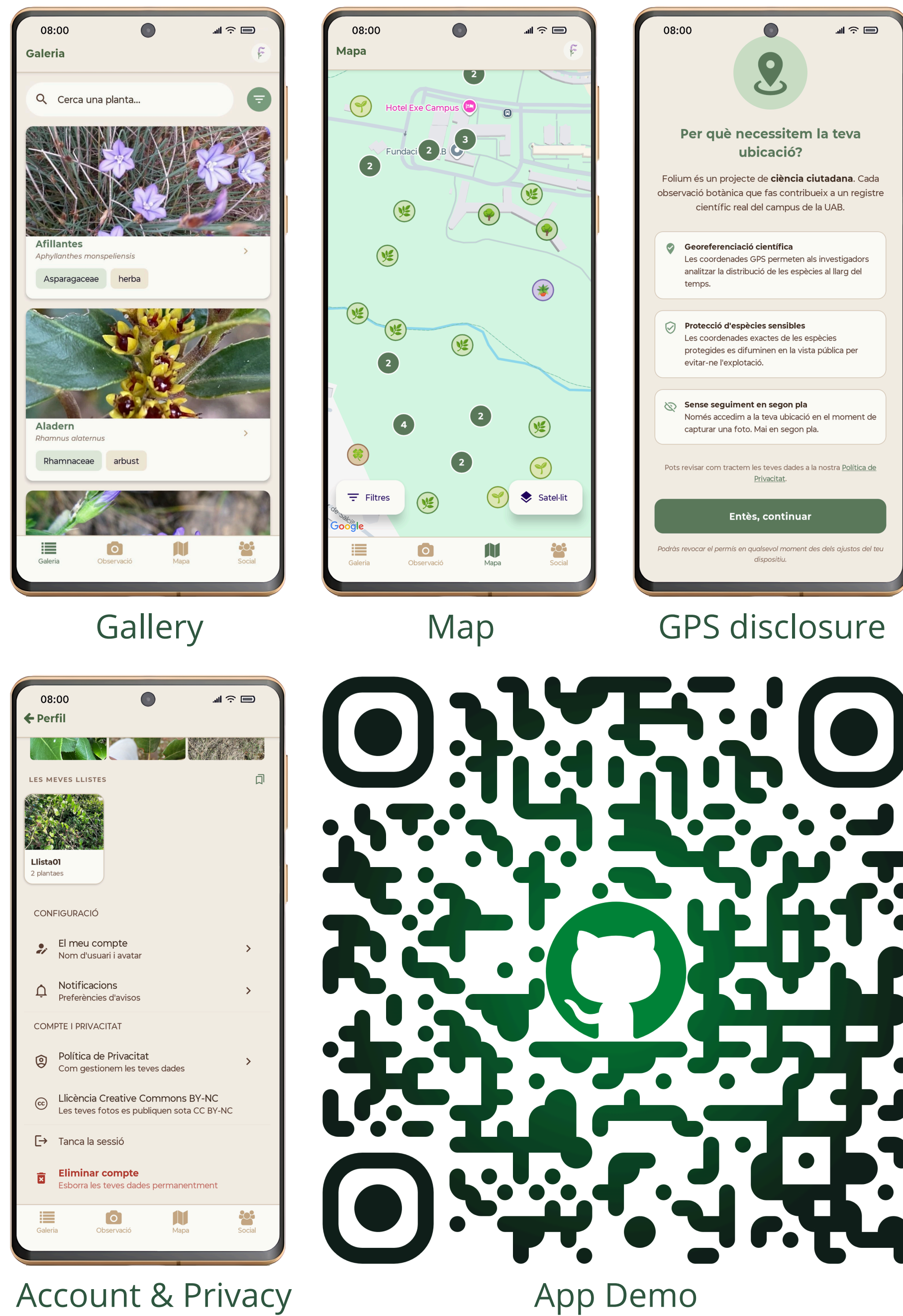
Validation Model



3 Results & Conclusions



App Screenshots



Key Achievements

- App published on **Google Play Store** (May 2026)
- Creative Commons BY-NC 4.0** licence for all UGC
- Consensus validation engine (iNaturalist-inspired, rule ≥ 50 pts)
- Push alerts to experts for anomalous observations (Edge Functions)
- Full GPS permission onboarding for fieldwork
- Total operating cost: **\$25 USD** (Google Play, one-time)



Conclusions

Folium demonstrates that a single developer in a university setting can deliver a **production-quality citizen-science platform**. The Living Lab model at UAB Campus is viable as a scalable approach for hyper-local biodiversity monitoring.

Open challenge: pilot data collection pending Play Store approval (June 2026).

Key Words

- Citizen Science
- Botanical Cataloging
- Supabase
- GDPR
- Mobile Deployment
- Biodiversity Monitoring
- React Native



References

- iNaturalist Research Grade criteria. [iNaturalist.org](https://www.inaturalist.org)
- Fastlane Documentation. docs.fastlane.tools
- Creative Commons. creativecommons.org
- Expo / React Native docs. docs.expo.dev
- Supabase documentation. supabase.com/docs

Future Work

- Expanded role hierarchy** — Finer-grained ranks for deeper community engagement and contribution incentives
- Project organizations** — Delimit geographic scope and species dataset per community or initiative
- Seasonal Wrapped** — Annual activity summaries per user to boost long-term retention